THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. During a solar eclipse, the Moon's shadow falls on the Earth. The region with total darkness is the: 1

(a) halo **(b)** penumbra **(c)** umbra **(d)** corona

2. A year in which the Earth makes exactly 366.25 days in an orbit would need: 1

(a) same **(b)** more frequent leap years

(c) no leap years **(d)** fewer leap years

3. A rainwater sample turns blue litmus red. The rain may be: 1

(a) salty **(b)** neutral **(c)** basic **(d)** acidic

4. A student sets up a tester with a cell, a bulb and two wires with free ends. The bulb glows when a steel spoon is placed across the ends. The spoon is: 1

(a) an insulator **(b)** a fuse **(c)** a conductor **(d)** a cell

5. Which metal is used to coat iron sheets to protect them from rusting? 1

(a) sulphur **(b)** zinc **(c)** coal **(d)** wax

6. A shop owner stores iron nails in a damp room. After some days, nails are covered with flaky reddish layer. The best advice to prevent this: 1

(a) store in dry place or oil them **(b)** store in water

(c) store in salt water **(d)** store in vinegar

7. A graph of height against age for a typical teenager shows: 1

(a) no change **(b)** a rapid rise during the growth spurt

(c) a fall **(d)** a straight line only

8. Which of the following shows a correct application of convection? 1

(a) sea breeze **(b)** spoon heating **(c)** cloth drying **(d)** sunlight on skin

9. A pendulum was timed for 10 oscillations: 18.2 s, 18.0 s, 18.4 s. The best estimate of T is: 1

(a) 1.8 s **(b)** 18.2 s **(c)** 0.55 s **(d)** 1.82 s

10. The reason that the human small intestine is long and coiled is: 1

(a) to make a noise

(b) to store food

(c) to digest water

(d) to provide a large surface area for absorption

11. A farmer notices that his legume plants in poor soil have swollen nodules in roots. This indicates: 1

(a) nitrogen-fixing bacteria **(b)** insects
(c) water logging **(d)** fungi attack

12. Why does the image in a mirror appear to be as far behind as the object is in front? 1

(a) mirror is thick
(b) mirror is smaller
(c) virtual image position follows the law of reflection
(d) light bends

13. A student claims that the Earth's shadow is formed only during lunar eclipses. The correct statement is: 1

(a) the shadow forms only then
(b) no shadow exists
(c) it is formed by the Moon
(d) the Earth always has a shadow but the Moon enters it only during eclipses

14. A liquid turns turmeric red. When liquid Y is added, the colour changes back to yellow. Liquid Y is: 1

(a) water **(b)** a base **(c)** sugar **(d)** an acid

15. A student connects five 1.5 V cells in series to a 6 V bulb. The bulb will: 1

(a) be dim **(b)** glow very bright and may fuse
(c) not glow **(d)** glow normally

16. A yellow brittle solid Y burns with a blue flame and its oxide turns blue litmus red. Y is: 1

(a) iron **(b)** sulphur **(c)** copper **(d)** magnesium

17. **Assertion (A):** Adolescence is a period of rapid growth in height and body changes. 1
Reason (R): Hormones secreted during puberty control these changes.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Heat always flows from a hotter body to a colder body. 1
Reason (R): Heat flows until the two bodies reach the same temperature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The SI unit of time is the second. 1
Reason (R): One minute is equal to 60 seconds.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Digestion of starch begins in the mouth. 1
Reason (R): Saliva contains salivary amylase, which breaks starch into sugar.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Calculate the speed of a train in km/h if it crosses a 300 m bridge in 30 s and its length is 200 m. 2

22. Explain why a person feels warm after eating food rich in carbohydrates. 2

23. Why does a plant in a closed jar with a burning candle and plant show that plants restore the air? 2

24. A ray is incident at 90° to a plane mirror. State the angle of incidence and the path of the reflected ray. 2

25. Explain why the Moon's light does not produce a clear shadow as strongly as the Sun's. 2

26. How can you distinguish between two colourless liquids, one acid and one base, with a single indicator? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Design an experiment to show that oxygen is released during photosynthesis, with controls. 3

28. A ray of light strikes a mirror with an incident angle of 55°. Find the angle between the incident ray and the reflected ray, and the angle between the mirror and the reflected ray. 3

29. The Earth rotates 360° in 24 h. Find how many degrees it rotates in 1 h and 4 min, and the time for 15° of rotation. 3

30. Explain why we use different indicators for different solutions. 3

31. Why should heavy appliances be connected through thick wiring and not thin wiring? 3

OR

Explain with equations how metals and non-metals form acidic and basic solutions with water. 3

32. Explain why a mixture of salt and sand is physically separated but a mixture of iron and sulphur on heating is not. 3

OR

A diabetic friend wants to join a picnic. What advice would you give about food and medicines? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Aditi helps her mother in the kitchen in Pune. The frying pan is made of aluminium but its handle is made of plastic. Aditi also notices that aluminium foil can be wrapped around parathas, and that the copper wire of the mixer is thin and long. 5

(i) Why is the pan made of a metal?

(ii) Why is the handle made of plastic?

(iii) Which property of aluminium allows it to be made into foil?

(iv) Name the property that allows copper to be made into wire and one more property that makes copper suitable for electric wires.

34. **Case study.** Neha, a Class 7 student in Nashik, prepares a hot saturated solution of copper sulphate in water, filters it and leaves it undisturbed to cool slowly overnight. Next morning, she finds beautiful blue crystals in the beaker. 5

(i) Name the process by which the crystals formed.

(ii) Is it a physical or a chemical change?

(iii) Why was a hot solution prepared?

(iv) Neha cooled another solution quickly and got only tiny crystals. Suggest how she can get larger crystals and why.

35. Explain the significance of soil acidity and the role of crops, lime, and organic manure in correcting it. 5

36. Explain why electrical appliances carry safety marks such as ISI and how to read the cell and fuse ratings. 5

OR

Explain how a student can show that metal oxides are basic and non-metal oxides acidic using an indicator. 5

37. Explain how to conduct an experiment to show that burning needs oxygen and produces carbon dioxide. 5

OR

Discuss the importance of respecting privacy, consent and safe behaviour for adolescents in society. 5

38. A group of students measure the temperature of black and white cloth in sunlight and shade over an hour. Predict and analyse the likely graph. 5

OR

A car's odometer reads 25 340 km at 9:00 a.m. and 25 420 km at 11:00 a.m. Its speedometer reads 60 km/h at 9:30 a.m. and 20 km/h at 10:30. Explain the difference between average and instantaneous speeds. 5

END OF QUESTION PAPER — Total: 80 Marks

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